Homework 2

Network Topology Design & Configuration

GNS3 Lab Report

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Introduction

This report documents the design, configuration, and verification of a multi-site enterprise network topology built in GNS3. The objective is to interconnect three isolated user groups (VLAN 10, 20, 30) through a central switching fabric, enforce security policies at Layers 3 and 4 via a FortiGate firewall and Cisco router ACLs, and provide controlled Internet access through NAT.

The network follows a collapsed-core design where a central distribution switch aggregates traffic from three access switches, each serving a single VLAN. All inter-VLAN routing and security filtering is handled by a FortiGate firewall acting as the default gateway for each subnet.

Topology Overview

Physical Topology

The topology consists of seven devices interconnected as shown in Table 1.

Table 1: Device Interconnection Table

From	**Port**	**To**	**Port**	**Purpose**
Cloud1 (Internet/NAT)	eth1 (VMnet8)	R1	Fa0/0	Internet access via DHCP
R1	Fa1/0	FortiGate	port1	WAN link (192.168.1.0/30)
FortiGate	port2	Central-Switch	Gi0/0	Main LAN trunk
Central-Switch	Gi0/1	Left-Switch	Gi0/0	Trunk (VLAN 10 only)
Central-Switch	Gi0/2	Middle-Switch	Gi0/0	Trunk (VLAN 20 only)
Central-Switch	Gi0/3	Right-Switch	Gi0/0	Trunk (VLAN 30 only)
Left-Switch	Gi0/1	PC1	eth0	Access port VLAN 10
Middle-Switch	Gi0/1	PC2	eth0	Access port VLAN 20
Right-Switch	Gi0/1	PC3	eth0	Access port VLAN 30

Logical Topology and VLAN Structure

The network is segmented into three VLANs, each corresponding to a distinct user group. The Central-Switch is the core of the Layer 2 design: it trunks all three VLANs to the FortiGate for routing, and distributes each VLAN to its dedicated access switch. Table 2 summarizes the VLAN assignment.

Table 2: VLAN Assignment

VLAN ID	Name	Subnet	Assigned To
10	USERS_A	10.10.10.0/24	PC1 (Left-Switch)
20	USERS_B	10.20.20.0/24	PC2 (Middle-Switch)
30	USERS_C	10.30.30.0/24	PC3 (Right-Switch)

Equipment & Software

- **Router R1:** Cisco C7200 (c7200-adventerprisek9-mz.152-4.M11)
- **Firewall:** FortiGate 7.0.9 VM
- **Switches:** Cisco IOSvL2 (vios_l2-adventerprisek9-m.03.2017)
- **End hosts:** VPCS (Virtual PC Simulator)
- **Cloud:** GNS3 Cloud node bridged to VMware VMnet8 (NAT)

Lab Environment

The GNS3 virtual machine runs on VMware NAT with the following environment parameters:

- **GNS3 VM network:** VMware NAT 10.0.0.0/24
- **Default gateway:** 10.0.0.2
- **DNS servers:** 8.8.8.8 and 8.8.4.4

The Cloud1 node in GNS3 is bridged to VMware VMnet8 and provides DHCP addressing to R1's Fa0/0 interface, enabling Internet access through the VMware NAT gateway.

IP Addressing Plan

Table 3 details every Layer 3 interface in the topology.

Table 3: IP Address Assignment

Device / Interface	IP Address	Method
R1 Fa0/0	DHCP (VMnet8)	DHCP from Cloud1
R1 Fa1/0	192.168.1.1/30	Static
FortiGate port1	192.168.1.2/30	Static
FortiGate port2.10	10.10.10.1/24	Static (GW for VLAN 10)
FortiGate port2.20	10.20.20.1/24	Static (GW for VLAN 20)
FortiGate port2.30	10.30.30.1/24	Static (GW for VLAN 30)
PC1	DHCP (10.10.10.10–100)	DHCP from FortiGate
PC2	DHCP (10.20.20.10–100)	DHCP from FortiGate
PC3	DHCP (10.30.30.10–100)	DHCP from FortiGate

Device Configurations

Router R1 (Cisco C7200)

R1 serves as the border router between the internal LAN and the Internet. It obtains a public (NAT) address on Fa0/0 via DHCP from Cloud1 (VMnet8), and provides a point-to-point WAN link to the FortiGate on Fa1/0. NAT overload is configured so that all internal subnets can reach the Internet through the single DHCP-obtained address. Two ACLs enforce Layer 3/4 restrictions:

- **ACL 10 (inbound on Fa1/0):** Blocks all traffic sourced from VLAN 30 (10.30.30.0/24) from entering the R1-FortiGate WAN link, effectively isolating VLAN 30 from the upstream network.
- **ACL 100 (outbound on Fa1/0):** Denies SSH (TCP/22) from VLAN 20 and VLAN 30 toward VLAN 10, preventing lateral SSH access to the USERS_A subnet.

Listing 1: R1 Configuration

```
hostname R1
no ip domain-lookup

interface FastEthernet0/0
 description WAN_toward_Cloud1_VMnet8
 ip address dhcp
 ip nat outside
 no shutdown

interface FastEthernet1/0
 description LAN_toward_FortiGate_port1
 ip address 192.168.1.1 255.255.255.252
 ip nat inside
 ip access-group 10 in
 ip access-group 100 out
 no shutdown

ip route 0.0.0.0 0.0.0.0 192.168.221.2

ip access-list standard NAT_SOURCES
 permit 192.168.1.0 0.0.0.3
 permit 10.10.10.0 0.0.0.255
 permit 10.20.20.0 0.0.0.255
 permit 10.30.30.0 0.0.0.255

ip nat inside source list NAT_SOURCES interface FastEthernet0/0 overload

access-list 10 deny   10.30.30.0 0.0.0.255
access-list 10 permit any

access-list 100 deny   tcp 10.20.20.0 0.0.0.255 10.10.10.0 0.0.0.255 eq
   22
access-list 100 deny   tcp 10.30.30.0 0.0.0.255 10.10.10.0 0.0.0.255 eq
   22
access-list 100 permit ip any any
```

FortiGate Firewall

The FortiGate acts as the central Layer 3 device for all internal VLANs. Each VLAN terminates on a subinterface of port2 (VLAN tagging via 802.1Q). The firewall provides DHCP services per subnet and enforces inter-VLAN security policies. On first boot after a factory reset, the admin password must be set via:

Listing 2: FortiGate Admin Password

```
config system admin
    edit admin
        set password YourSecurePass123
    next
end
```

Six firewall policies are defined, ordered from most specific to least:

1. **VLAN10 to Internet:** Allow with NAT.
2. **VLAN20 to Internet:** Allow with NAT.
3. **Block VLAN30 Internet:** Deny all outbound from VLAN 30 to port1.
4. **Block SSH between VLANs:** Deny SSH traffic from any internal subnet to any other internal subnet.
5. **Allow Inter-VLAN:** Permit all remaining traffic between internal subnets.
6. **Block Internet In:** Deny all inbound traffic from the WAN interface.

Listing 3: FortiGate Configuration: Interfaces and Routes

```
config system interface
    edit "port1"
        set mode static
        set ip 192.168.1.2 255.255.255.252
        set allowaccess ping https ssh
        set role wan
        set description "WAN_toward_R1"
    next
    edit "port2"
        set mode static
        set ip 0.0.0.0 0.0.0.0
        set allowaccess ping
        set role lan
        set description "LAN_Trunk_to_Central-Switch"
    next
    edit "port2.10"
        set vdom "root"
        set interface "port2"
        set vlanid 10
        set mode static
        set ip 10.10.10.1 255.255.255.0
        set allowaccess ping
        set role lan
        set description "VLAN10_USERS_A"
    next
```

```
    edit "port2.20"
        set vdom "root"
        set interface "port2"
        set vlanid 20
        set mode static
        set ip 10.20.20.1 255.255.255.0
        set allowaccess ping
        set role lan
        set description "VLAN20_USERS_B"
    next
    edit "port2.30"
        set vdom "root"
        set interface "port2"
        set vlanid 30
        set mode static
        set ip 10.30.30.1 255.255.255.0
        set allowaccess ping
        set role lan
        set description "VLAN30_USERS_C"
    next
end

config router static
    edit 1
        set dst 0.0.0.0 0.0.0.0
        set gateway 192.168.1.1
        set device "port1"
    next
end
```

Central-Switch

The Central-Switch is the distribution layer. It trunks all three VLANs toward the FortiGate and distributes each VLAN on a dedicated trunk port to its respective access switch. VLAN pruning is applied to each trunk so that only the necessary VLANs traverse each link.

Listing 4: Central-Switch Configuration

```
hostname Central-Switch
vlan 10
 name USERS_A
vlan 20
 name USERS_B
vlan 30
 name USERS_C

interface GigabitEthernet0/0
 description TRUNK_to_FortiGate_port2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30

interface GigabitEthernet0/1
 description TRUNK_to_Left-Switch
 switchport trunk encapsulation dot1q
```

```
 switchport mode trunk
 switchport trunk allowed vlan 10

interface GigabitEthernet0/2
 description TRUNK_to_Middle-Switch
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20

interface GigabitEthernet0/3
 description TRUNK_to_Right-Switch
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 30
```

Access Switches (Left / Middle / Right)

Each access switch is responsible for a single VLAN. The trunk toward the Central-Switch is restricted to that VLAN only. End-host ports are configured as access ports with PortFast enabled for immediate forwarding. The configuration is identical in structure across all three switches; only the VLAN ID and names differ.

Listing 5: Left-Switch Configuration (VLAN 10)

```
hostname Left-Switch
vlan 10
 name USERS_A

interface GigabitEthernet0/0
 description TRUNK_to_Central-Switch
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10

interface GigabitEthernet0/1
 description ACCESS_PC1_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
```

The Middle-Switch and Right-Switch follow the same pattern with VLAN 20 (USERS_B) and VLAN 30 (USERS_C) respectively. The end-host port is Gi0/1 in each case.

End Hosts (VPCS)

All three PCs obtain their IP configuration via DHCP. No static configuration is needed on the hosts themselves.

Listing 6: PC Configuration

```
PC1> ip dhcp
PC2> ip dhcp
PC3> ip dhcp
```

Security Policy Summary

The security model combines Cisco ACLs (router-side) and FortiGate firewall policies. Table 4 gives the complete policy matrix.

Table 4: Security Policy Matrix

Traffic Flow	ICMP	HTTP	SSH	Other	Enforcement
VLAN 10 → Internet	Allow	Allow	Allow	Allow	R1 NAT + FG Policy 1
VLAN 20 → Internet	Allow	Allow	Allow	Allow	R1 NAT + FG Policy 2
VLAN 30 → Internet	Deny	Deny	Deny	Deny	FG Policy 3
VLAN 10 VLAN 20	Allow	Allow	Deny	Allow	FG Policies 4-5
VLAN 10 VLAN 30	Allow	Allow	Deny	Allow	FG Policies 4-5
VLAN 20 VLAN 30	Allow	Allow	Deny	Allow	FG Policies 4-5
VLAN 30 → R1 Fa1/0 (inbound)	Deny	Deny	Deny	Deny	R1 ACL 10
Internet → Internal	Deny	Deny	Deny	Deny	FG Policy 6

The layered approach ensures defense in depth:

- The **FortiGate** blocks VLAN 30 Internet access at the firewall level (Policy 3) and blocks all inbound WAN traffic (Policy 6).
- **R1 ACL 10** provides a second barrier: even if a FortiGate misconfiguration were to pass VLAN 30 traffic upstream, R1 drops it at the ingress of the WAN link.
- SSH is blocked unconditionally between all VLAN pairs, preventing lateral movement via SSH. This is enforced by FortiGate Policy 4 and redundantly by R1 ACL 100 for traffic that reaches the router.

DHCP Configuration

Three DHCP scopes are configured on the FortiGate, one per VLAN subinterface:

Table 5: DHCP Scopes

Scope	Interface	Range	Gateway	DNS
1	port2.10 (VLAN 10)	10.10.10.10 – 10.10.10.100	10.10.10.1	8.8.8.8, 8.8.4.4
2	port2.20 (VLAN 20)	10.20.20.10 – 10.20.20.100	10.20.20.1	8.8.8.8, 8.8.4.4
3	port2.30 (VLAN 30)	10.30.30.10 – 10.30.30.100	10.30.30.1	8.8.8.8, 8.8.4.4

Verification Plan

Table 6 lists the expected results for each test case, confirming the correctness of both connectivity and security enforcement.

Table 6: Verification Test Matrix

Source	Destination	Expected	Reason
PC1	8.8.8.8	Success	VLAN 10 has full Internet access
PC2	8.8.8.8	Success	VLAN 20 has full Internet access
PC3	8.8.8.8	Failure	VLAN 30 Internet blocked (FG Policy 3)
PC1	10.20.20.10	Success	Inter-VLAN routing allowed (FG Policy 5)
PC1	10.30.30.10	Success	Inter-VLAN routing allowed (FG Policy 5)
PC2	ssh 10.10.10.10	Failure	SSH inter-VLAN blocked (FG Policy 4, R1 ACL 100)
PC3	ssh 10.10.10.10	Failure	SSH inter-VLAN blocked (FG Policy 4, R1 ACL 100)
PC3	192.168.1.1	Failure	VLAN 30 blocked on R1 Fa1/0 (ACL 10)

The main troubleshooting procedures encountered during the lab included:

- **PC1 DHCP failure:** Caused by the VLAN 10 entry missing from the Left-Switch VLAN database after a factory reset. Resolved by re-creating `vlan 10` and re-assigning the access port. The fix was verified with `show vlan brief`, which then showed VLAN 10 active with Gi0/1 listed under it.
- **Internet unreachable (ICMP type 3, code 1 from R1):** The default route on R1 pointed to the wrong VMnet8 gateway. The correct gateway (`.2` of the VMnet8 subnet) was obtained from the Windows host's `ipconfig` output and applied to R1. The command used was `no ip route 0.0.0.0 0.0.0.0` followed by `ip route 0.0.0.0 0.0.0.0 <correct-gateway>`.
- **R1 Fa0/0 shows no IP (unassigned):** Caused by Cloud1 not providing DHCP. Checked in GNS3 that Cloud1 was using eth1/VMnet8 (not eth0/VMnet0), and that was the problem. Switched it to eth1

Conclusion

This lab demonstrated a complete enterprise network deployment comprising VLAN segmentation, trunking with pruning, inter-VLAN routing via a firewall, NAT with overload, DHCP per subnet, and multi-layer security enforcement. The use of FortiGate for both routing and stateful inspection, combined with Cisco ACLs at the border, provides a realistic defense-in-depth posture suitable for a segmented corporate LAN. All configurations were validated against the test matrix defined in Section 7.

Screenshots

This appendix contains screenshots taken during the lab, organized in the same order as the configuration and verification steps described in the report.

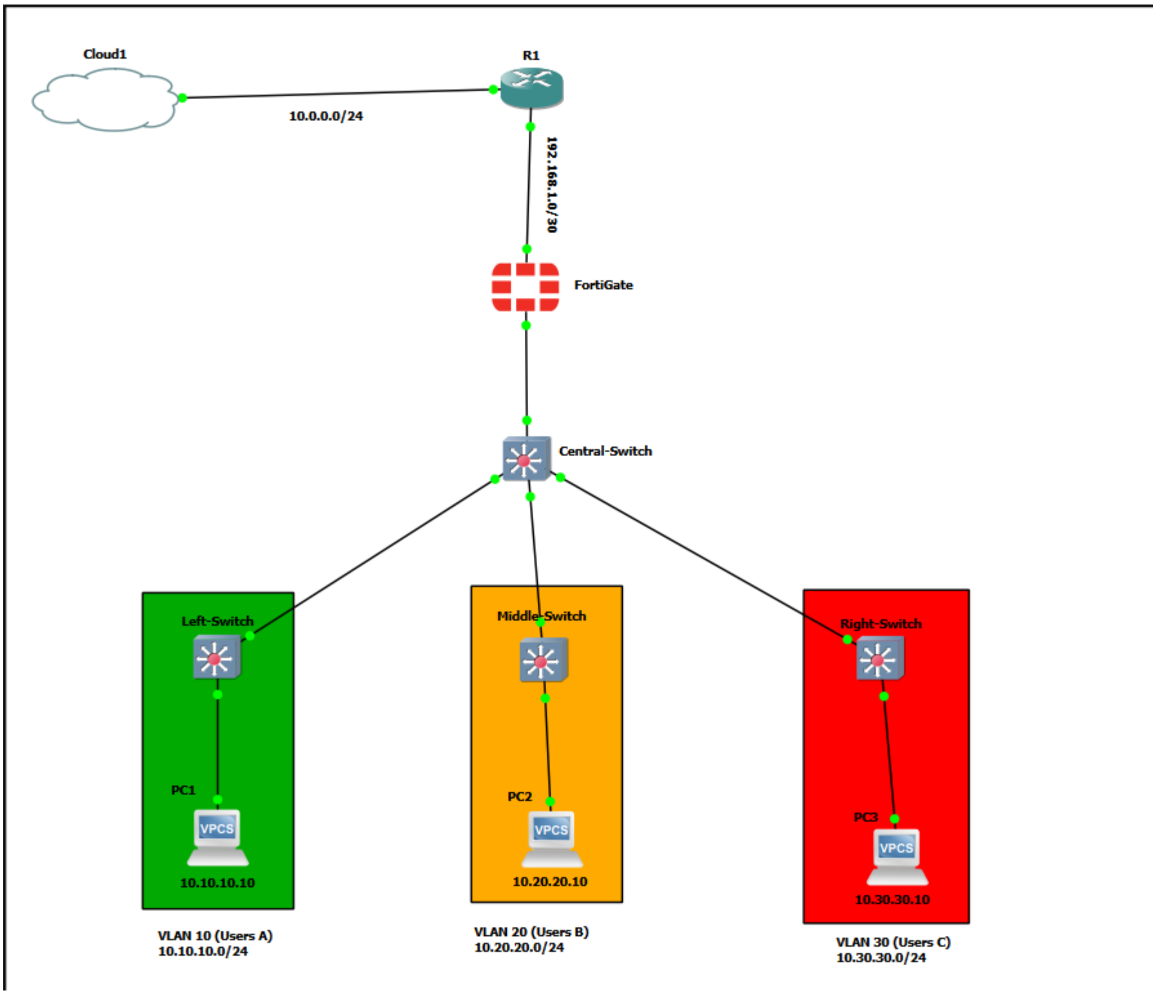


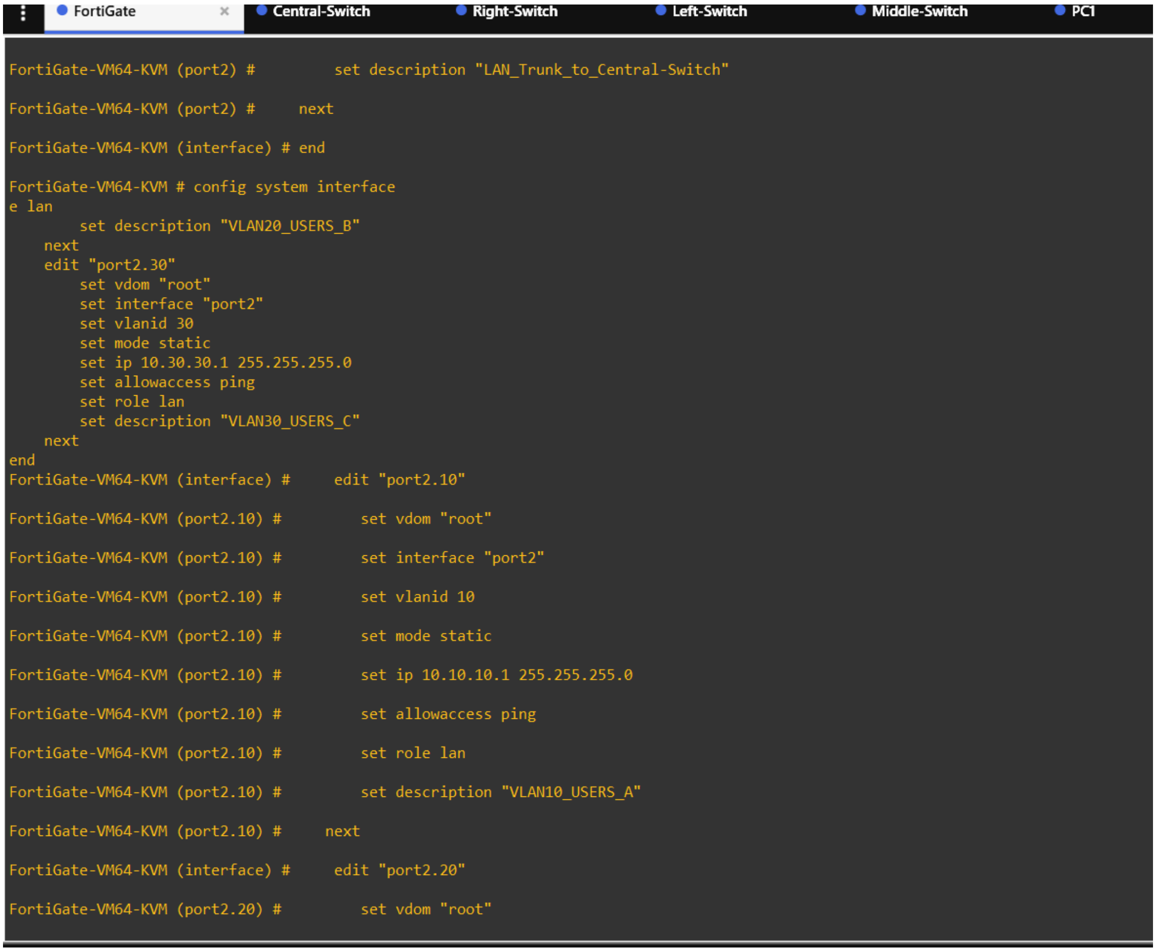
Figure 1: TOPOLOGY

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#
R1(config)#interface FastEthernet0/0
R1(config-if)# description WAN_toward_Cloud1_VMnet8
R1(config-if)# ip address dhcp
R1(config-if)# ip nat outside
R1(config-if)# no shutdown
R1(config-if)#
R1(config-if)#interface FastEthernet1/0
R1(config-if)# description LAN_toward_FortiGate_port1
R1(config-if)# ip address 192.168.1.1 255.255.255.252
R1(config-if)# ip nat inside
R1(config-if)# ip access-group 10 in
R1(config-if)# ip access-group 100 out
R1(config-if)# no shutdown
R1(config-if)#
R1(config-if)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
R1(config)#
R1(config)#ip access-list standard NAT_SOURCES
R1(config-std-nacl)# permit 192.168.1.0 0.0.0.3
R1(config-std-nacl)# permit 10.10.10.0 0.0.0.255
R1(config-std-nacl)# permit 10.20.20.0 0.0.0.255
R1(config-std-nacl)# permit 10.30.30.0 0.0.0.255
R1(config-std-nacl)#
R1(config-std-nacl)#$list NAT_SOURCES interface FastEthernet0/0 overload
R1(config)#
R1(config)#access-list 10 deny   10.30.30.0 0.0.0.255
R1(config)#access-list 10 permit any
R1(config)#
R1(config)#$ 100 deny   tcp 10.20.20.0 0.0.0.255 10.10.10.0 0.0.0.255 eq 22
R1(config)#$ 100 deny   tcp 10.30.30.0 0.0.0.255 10.10.10.0 0.0.0.255 eq 22
R1(config)#access-list 100 permit ip any any
R1(config)#
R1(config)#end
R1#write memory
```

Figure 2: ROUTER

```
System is starting...
Starting system maintenance...
Serial number is FGVMEVBEBRTMNR97


FortiGate-VM64-KVMadminn:
Password:
Login incorrect


FortiGate-VM64-KVM login: admin
Password:
Welcome!

WARNING: File System Check Recommended! An unsafe reboot may have caused an inconsistency in the disk drive.
It is strongly recommended that you check the file system consistency before proceeding.
Please run 'execute disk list' and then 'execute disk scan <ref#>'.
Note: The device will reboot and scan the disk during startup. This may take up to an hour.
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # config system interface

FortiGate-VM64-KVM (interface) #     edit "port1"

FortiGate-VM64-KVM (port1) #         set mode static

FortiGate-VM64-KVM (port1) #         set ip 192.168.1.2 255.255.255.252

FortiGate-VM64-KVM (port1) #         set allowaccess ping https ssh

FortiGate-VM64-KVM (port1) #         set role wan

FortiGate-VM64-KVM (port1) #         set description "WAN_toward_R1"

FortiGate-VM64-KVM (port1) #     next

FortiGate-VM64-KVM (interface) #     edit "port2"

FortiGate-VM64-KVM (port2) #         set mode static

FortiGate-VM64-KVM (port2) #         set ip 0.0.0.0 0.0.0.0

FortiGate-VM64-KVM (port2) #         set allowaccess ping

FortiGate-VM64-KVM (port2) #         set role lan
```

Figure 3: FORTIGATE Interfaces Config




```
FortiGate-VM64-KVM (port2) #         set description "LAN_Trunk_to_Central-Switch"

FortiGate-VM64-KVM (port2) #     next

FortiGate-VM64-KVM (interface) # end

FortiGate-VM64-KVM # config system interface
e lan
        set description "VLAN20_USERS_B"
    next
    edit "port2.30"
        set vdom "root"
        set interface "port2"
        set vlanid 30
        set mode static
        set ip 10.30.30.1 255.255.255.0
        set allowaccess ping
        set role lan
        set description "VLAN30_USERS_C"
    next
end
FortiGate-VM64-KVM (interface) #     edit "port2.10"

FortiGate-VM64-KVM (port2.10) #         set vdom "root"

FortiGate-VM64-KVM (port2.10) #         set interface "port2"

FortiGate-VM64-KVM (port2.10) #         set vlanid 10

FortiGate-VM64-KVM (port2.10) #         set mode static

FortiGate-VM64-KVM (port2.10) #         set ip 10.10.10.1 255.255.255.0

FortiGate-VM64-KVM (port2.10) #         set allowaccess ping

FortiGate-VM64-KVM (port2.10) #         set role lan

FortiGate-VM64-KVM (port2.10) #         set description "VLAN10_USERS_A"

FortiGate-VM64-KVM (port2.10) #     next

FortiGate-VM64-KVM (interface) #     edit "port2.20"

FortiGate-VM64-KVM (port2.20) #         set vdom "root"
```

Figure 4: FORTIGATE Interfaces Config

```
FortiGate-VM64-KVM (1) #         set device "port1"

FortiGate-VM64-KVM (1) #     next

FortiGate-VM64-KVM (static) # end

FortiGate-VM64-KVM # config system dhcp server

FortiGate-VM64-KVM (server) # set netmask 255.255.255.0
        set dns-server1 8.8.8.8
        set dns-server2 8.8.4.4
        config ip-range
            edit 1
                set start-ip 10.30.30.10
                set end-ip   10.30.30.100
            next
        end
    next
end    edit 1

FortiGate-VM64-KVM (1) #         set status enable

FortiGate-VM64-KVM (1) #         set interface "port2.10"

FortiGate-VM64-KVM (1) #         set default-gateway 10.10.10.1

FortiGate-VM64-KVM (1) #         set netmask 255.255.255.0

FortiGate-VM64-KVM (1) #         set dns-server1 8.8.8.8

FortiGate-VM64-KVM (1) #         set dns-server2 8.8.4.4

FortiGate-VM64-KVM (1) #         config ip-range

FortiGate-VM64-KVM (ip-range) #            edit 1

FortiGate-VM64-KVM (1) #                 set start-ip 10.10.10.10

FortiGate-VM64-KVM (1) #                 set end-ip   10.10.10.100

FortiGate-VM64-KVM (1) #             next

FortiGate-VM64-KVM (ip-range) #         end

FortiGate-VM64-KVM (1) #     next
```

Figure 5: FORTIGATE DHCP Config

```
FortiGate-VM64-KVM # config firewall policy
 name "VLAN20_to_Internet"
        set srcintf "port2.20"
        set dstintf "port1"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set nat enable
        set logtraffic all
    next
    edit 3
        set name "BLOCK_VLAN30_INTERNET"
        set srcintf "port2.30"
        set dstintf "port1"
        set srcaddr "all"
        set dstaddr "all"
        set action deny
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
    edit 4
        set name "BLOCK_SSH_BETWEEN_VLANS"
        set srcintf "port2.10" "port2.20" "port2.30"
        set dstintf "port2.10" "port2.20" "port2.30"
        set srcaddr "all"
        set dstaddr "all"
        set action deny
        set schedule "always"
        set service "SSH"
        set logtraffic all
    next
    edit 5
        set name "ALLOW_INTER_VLAN"
        set srcintf "port2.10" "port2.20" "port2.30"
        set dstintf "port2.10" "port2.20" "port2.30"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
```

Figure 6: FORTIGATE FIREWALL Config

```
        set name "ALLOW_INTER_VLAN"
        set srcintf "port2.10" "port2.20" "port2.30"
        set dstintf "port2.10" "port2.20" "port2.30"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
    edit 6
        set name "BLOCK_INTERNET_IN"
        set srcintf "port1"
        set dstintf "port2.10" "port2.20" "port2.30"
        set srcaddr "all"
        set dstaddr "all"
        set action deny
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
end
FortiGate-VM64-KVM (policy) #     edit 1

FortiGate-VM64-KVM (1) #         set name "VLAN10_to_Internet"

FortiGate-VM64-KVM (1) #         set srcintf "port2.10"

FortiGate-VM64-KVM (1) #         set dstintf "port1"

FortiGate-VM64-KVM (1) #         set srcaddr "all"

FortiGate-VM64-KVM (1) #         set dstaddr "all"

FortiGate-VM64-KVM (1) #         set action accept

FortiGate-VM64-KVM (1) #         set schedule "always"

FortiGate-VM64-KVM (1) #         set service "ALL"

FortiGate-VM64-KVM (1) #         set nat enable

FortiGate-VM64-KVM (1) #         set logtraffic all

FortiGate-VM64-KVM (1) #     next
```

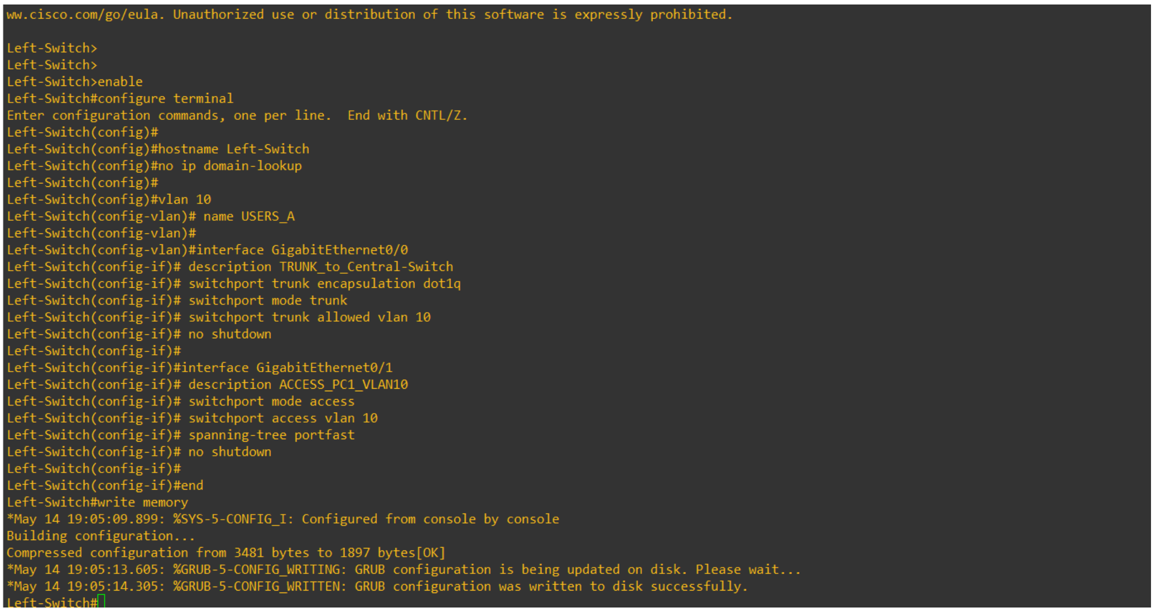
Figure 7: FORTIGATE FIREWALL Config

```
FortiGate-VM64-KVM (policy) #     edit 4
FortiGate-VM64-KVM (4) #        set name "BLOCK_SSH_BETWEEN_VLANS"
FortiGate-VM64-KVM (4) #        set srcintf "port2.10" "port2.20" "port2.30"
FortiGate-VM64-KVM (4) #        set dstintf "port2.10" "port2.20" "port2.30"
FortiGate-VM64-KVM (4) #        set srcaddr "all"
FortiGate-VM64-KVM (4) #        set dstaddr "all"
FortiGate-VM64-KVM (4) #        set action deny
FortiGate-VM64-KVM (4) #        set schedule "always"
FortiGate-VM64-KVM (4) #        set service "SSH"
FortiGate-VM64-KVM (4) #        set logtraffic all
FortiGate-VM64-KVM (4) #    next
FortiGate-VM64-KVM (policy) #     edit 5
FortiGate-VM64-KVM (5) #        set name "ALLOW_INTER_VLAN"
FortiGate-VM64-KVM (5) #        set srcintf "port2.10" "port2.20" "port2.30"
FortiGate-VM64-KVM (5) #        set dstintf "port2.10" "port2.20" "port2.30"
FortiGate-VM64-KVM (5) #        set srcaddr "all"
FortiGate-VM64-KVM (5) #        set dstaddr "all"
FortiGate-VM64-KVM (5) #        set action accept
FortiGate-VM64-KVM (5) #        set schedule "always"
FortiGate-VM64-KVM (5) #        set service "ALL"
FortiGate-VM64-KVM (5) #        set logtraffic all
FortiGate-VM64-KVM (5) #    next
FortiGate-VM64-KVM (policy) #     edit 6
```

Figure 8: FORTIGATE FIREWALL Config

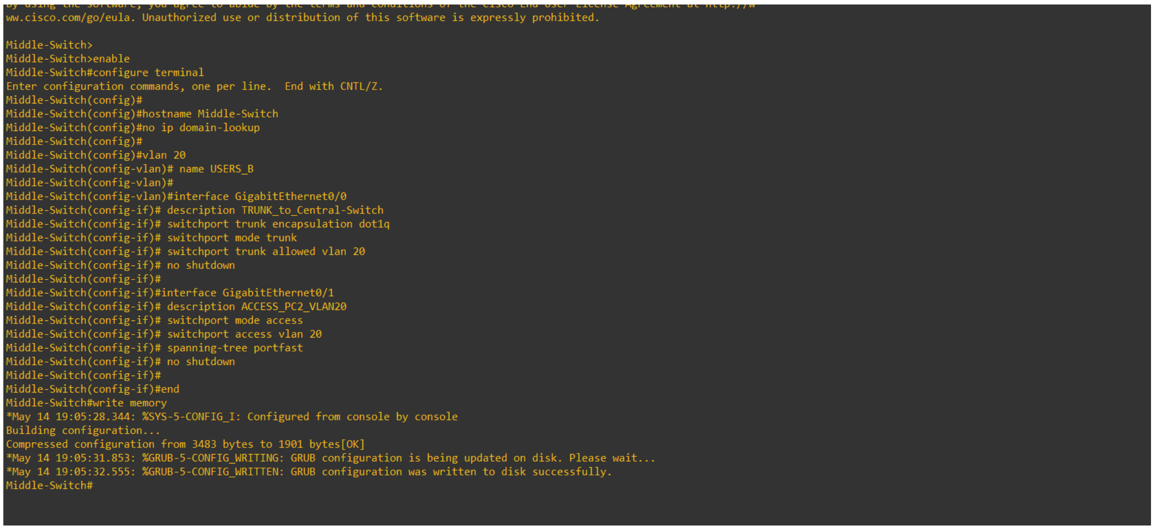
```
Central-Switch>enable
Central-Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Central-Switch(config)#
Central-Switch(config)#hostname Central-Switch
Central-Switch(config)#no ip domain-lookup
Central-Switch(config)#
Central-Switch(config)#vlan 10
Central-Switch(config-vlan)# name USERS_A
Central-Switch(config-vlan)#vlan 20
Central-Switch(config-vlan)# name USERS_B
Central-Switch(config-vlan)#vlan 30
Central-Switch(config-vlan)# name USERS_C
Central-Switch(config-vlan)#
Central-Switch(config-vlan)#interface GigabitEthernet0/0
Central-Switch(config-if)# description TRUNK_to_FortiGate_port2
Central-Switch(config-if)# switchport trunk encapsulation dot1q
Central-Switch(config-if)# switchport mode trunk
Central-Switch(config-if)# switchport trunk allowed vlan 10,20,30
Central-Switch(config-if)# no shutdown
Central-Switch(config-if)#
Central-Switch(config-if)#interface GigabitEthernet0/1
Central-Switch(config-if)# description TRUNK_to_Left-Switch
Central-Switch(config-if)# switchport trunk encapsulation dot1q
Central-Switch(config-if)# switchport mode trunk
Central-Switch(config-if)# switchport trunk allowed vlan 10
Central-Switch(config-if)# no shutdown
Central-Switch(config-if)#
Central-Switch(config-if)#interface GigabitEthernet0/2
Central-Switch(config-if)# description TRUNK_to_Middle-Switch
Central-Switch(config-if)# switchport trunk encapsulation dot1q
Central-Switch(config-if)# switchport mode trunk
Central-Switch(config-if)# switchport trunk allowed vlan 20
Central-Switch(config-if)# no shutdown
Central-Switch(config-if)#
Central-Switch(config-if)#interface GigabitEthernet0/3
Central-Switch(config-if)# description TRUNK_to_Right-Switch
Central-Switch(config-if)# switchport trunk encapsulation dot1q
Central-Switch(config-if)# switchport mode trunk
Central-Switch(config-if)# switchport trunk allowed vlan 30
Central-Switch(config-if)# no shutdown
Central-Switch(config-if)#
Central-Switch(config-if)#end
Central-Switch#write memory
*May 14 19:04:03.368: %SYS-5-CONFIG_I: Configured from console by console
```

Figure 9: CENTRAL SWITCH



```
ww.cisco.com/go/eula. Unauthorized use or distribution of this software is expressly prohibited.

Left-Switch>
Left-Switch>
Left-Switch>enable
Left-Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Left-Switch(config)#
Left-Switch(config)#hostname Left-Switch
Left-Switch(config)#no ip domain-lookup
Left-Switch(config)#
Left-Switch(config)#vlan 10
Left-Switch(config-vlan)# name USERS_A
Left-Switch(config-vlan)#
Left-Switch(config-vlan)#interface GigabitEthernet0/0
Left-Switch(config-if)# description TRUNK_to_Central-Switch
Left-Switch(config-if)# switchport trunk encapsulation dot1q
Left-Switch(config-if)# switchport mode trunk
Left-Switch(config-if)# switchport trunk allowed vlan 10
Left-Switch(config-if)# no shutdown
Left-Switch(config-if)#
Left-Switch(config-if)#interface GigabitEthernet0/1
Left-Switch(config-if)# description ACCESS_PC1_VLAN10
Left-Switch(config-if)# switchport mode access
Left-Switch(config-if)# switchport access vlan 10
Left-Switch(config-if)# spanning-tree portfast
Left-Switch(config-if)# no shutdown
Left-Switch(config-if)#
Left-Switch(config-if)#end
Left-Switch#write memory
*May 14 19:05:09.899: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
Compressed configuration from 3481 bytes to 1897 bytes[OK]
*May 14 19:05:13.605: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*May 14 19:05:14.305: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Left-Switch#
```

Figure 10: LEFT SWITCH



```
ww.cisco.com/go/eula. Unauthorized use or distribution of this software is expressly prohibited.

Middle-Switch>
Middle-Switch>enable
Middle-Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Middle-Switch(config)#
Middle-Switch(config)#hostname Middle-Switch
Middle-Switch(config)#no ip domain-lookup
Middle-Switch(config)#
Middle-Switch(config)#vlan 20
Middle-Switch(config-vlan)# name USERS_B
Middle-Switch(config-vlan)#
Middle-Switch(config-vlan)#interface GigabitEthernet0/0
Middle-Switch(config-if)# description TRUNK_to_Central-Switch
Middle-Switch(config-if)# switchport trunk encapsulation dot1q
Middle-Switch(config-if)# switchport mode trunk
Middle-Switch(config-if)# switchport trunk allowed vlan 20
Middle-Switch(config-if)# no shutdown
Middle-Switch(config-if)#
Middle-Switch(config-if)#interface GigabitEthernet0/1
Middle-Switch(config-if)# description ACCESS_PC2_VLAN20
Middle-Switch(config-if)# switchport mode access
Middle-Switch(config-if)# switchport access vlan 20
Middle-Switch(config-if)# spanning-tree portfast
Middle-Switch(config-if)# no shutdown
Middle-Switch(config-if)#
Middle-Switch(config-if)#end
Middle-Switch#write memory
*May 14 19:05:28.344: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
Compressed configuration from 3483 bytes to 1901 bytes[OK]
*May 14 19:05:31.853: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*May 14 19:05:32.555: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Middle-Switch#
```

Figure 11: MIDDLE SWITCH

```
Left-Switch#write erase
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Left-Switch#
*May 14 19:05:37.547: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Left-Switch#enable
Left-Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Left-Switch(config)#
Left-Switch(config)#hostname Right-Switch
Right-Switch(config)#no ip domain-lookup
Right-Switch(config)#
Right-Switch(config)#vlan 30
Right-Switch(config-vlan)# name USERS_C
Right-Switch(config-vlan)#
Right-Switch(config-vlan)#interface GigabitEthernet0/0
Right-Switch(config-if)# description TRUNK_to_Central-Switch
Right-Switch(config-if)# switchport trunk encapsulation dot1q
Right-Switch(config-if)# switchport mode trunk
Right-Switch(config-if)# switchport trunk allowed vlan 30
Right-Switch(config-if)# no shutdown
Right-Switch(config-if)#
Right-Switch(config-if)#interface GigabitEthernet0/1
Right-Switch(config-if)# description ACCESS_PC3_VLAN30
Right-Switch(config-if)# switchport mode access
Right-Switch(config-if)# switchport access vlan 30
Right-Switch(config-if)# spanning-tree portfast
Right-Switch(config-if)# no shutdown
Right-Switch(config-if)#
Right-Switch(config-if)#end
Right-Switch#write memory
Building configuration...

*May 14 19:05:51.697: %SYS-5-CONFIG_I: Configured from console by consoleCompressed configuration from 3482 bytes to 1906
```

Figure 12: RIGHT SWITCH

```
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file


PC1> ip dhcp
DDORA IP 10.10.10.10/24 GW 10.10.10.1
```

Figure 13: PC1

```
FortiGate  Central-Switch  Right-Switch  Left-Switch  Middle-Switch  PC1  PC2  PC3

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file


PC2> ip dhcp
DDORA IP 10.20.20.10/24 GW 10.20.20.1

PC2> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=126 time=97.236 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=126 time=112.322 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=126 time=109.935 ms
^C
PC2>
```

Figure 14: PC2

Figure 15: PC3

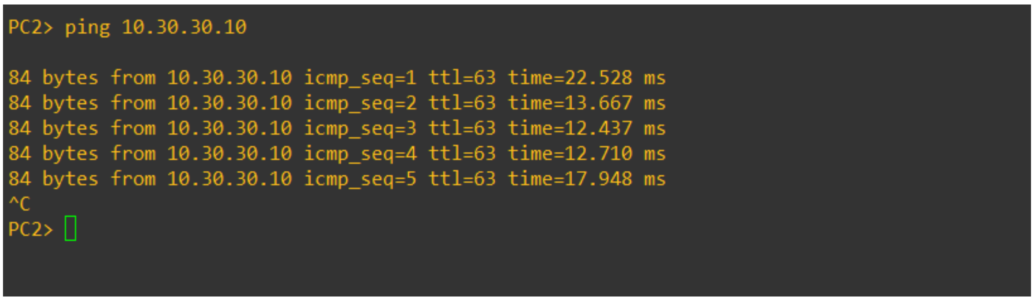

Figure 16: INTER VLAN ROUTING Test